

CHEATSHEET · QUICK REFERENCE

Lab 28 — Troubleshooting Methodology Walk-Through

Break nginx and apply the Server+ seven-step model end-to-end: identify, theorise, test, plan, implement, verify, RCA and document.

The methodology steps

Step	Action
1	Identify problem + scope (question, backup first)
2	Establish a theory (question the obvious)
3	Test the theory
4	Plan of action (notify users)
5	Implement or escalate (one change at a time)
6	Verify full functionality + preventive measures
7	Root cause analysis
8	Document throughout

Evidence-gathering commands

```
journalctl -u nginx -n 40 --no-pager
Service logs

systemctl status nginx
Service state

ss -tlnp 'sport = :22'
Prove a port conflict

cp /etc/nginx/sites-enabled/default /root/default.pre-fix
Back up before changes

nginx -t && systemctl restart nginx
Validate then apply the fix
```

RCA (5-whys) template

Why	Answer
nginx failed	Listen port conflict with sshd
port changed	Copy-paste from a sandbox config
bad config reached prod	No CI check on config edits
no CI check	No pipeline ownership defined
no ownership	Role unfilled after re-org